EXPERIMENT :INTERACTIVE STUDENT REGISTRATION FORM WITH CSS ANIMATIONS AND HTML5 CLIENT-SIDE VALIDATION

Aim:

To design and develop an interactive student registration form using HTML5 and CSS3, incorporating form validation, CSS animations, input focus effects, and responsive styling.

PROCEDURE

1. Open Notepad and create a new HTML file named `registration.html`.
2. Write the basic HTML5 structure and set the page title as **Student Registration Form**.
3. Create a registration form with fields for Full Name, Email Address, Password, Confirm Password, Mobile Number, Date of Birth, Gender, Department, and Address.
4. Use HTML5 input elements such as `text`, `email`, `password`, `tel`, and `date` to collect student details.
5. Apply validation attributes such as `required`, `minlength`, `maxlength`, and `pattern` to validate the user inputs.
6. Add Register and Reset buttons to submit the form and clear the entered details.
7. Create a separate CSS file named `regstyle.css` and link it to the HTML document using the `<link>` tag.
8. Apply CSS styling to set the background gradient, center the registration form, and add padding, rounded corners, and box shadows.
9. Use CSS `@keyframes` to animate the form when the page loads.
10. Apply the `:focus` selector and `transition` property to create smooth visual effects on input fields.
11. Use the `:hover` selector to change the appearance of the Register and Reset buttons when the mouse pointer moves over them.
12. Apply media queries to make the registration form responsive on different screen sizes.
13. Save both files in the same folder and open `registration.html` in a web browser.
14. Test the form by submitting empty fields, entering invalid email addresses, checking password length, testing mobile number validation, and using the Reset button.
15. Verify that the form displays browser validation messages and that the CSS animations and interactive effects work correctly.

CODE

```
<!-- Full Name -->
<label for="name">Full Name</label>
<input type="text" id="name" name="name"
      placeholder="Enter your full name"
      minlength="3" required>

<!-- Email -->
<label for="email">Email Address</label>
<input type="email" id="email" name="email"
      placeholder="Enter your email" required>
          <br> <br>

<!-- Password -->
<label for="password">Password</label>
<input type="password" id="password" name="password"
      placeholder="Minimum 8 characters"
      minlength="8" required>
          <br> <br>


<!-- Confirm Password -->
<label for="confirm-password">Confirm Password</label>
<input type="password" id="confirm-password"
      name="confirm-password"
      placeholder="Re-enter your password" required>
          <br> <br>


<!-- Mobile Number -->
<label for="mobile">Mobile Number</label>
<input type="tel" id="mobile" name="mobile"
      placeholder="Enter 10-digit mobile number"
      pattern="[0-9]{10}" maxlength="10" required>
          <br> <br>


<!-- Date of Birth -->
<label for="dob">Date of Birth</label>
<input type="date" id="dob" name="dob" required>
          <br> <br>



<!DOCTYPE html>
```

```
<html>
<head>
   <title>Student Registration Form</title>
   <link rel="stylesheet" href="regstyle.css">
</head>

<body>

   <div class="form-container">

      <h1>Student Registration</h1>
      <p>Fill in your details to register</p>

      <form action="#" method="get">
         <!-- Gender -->
         <label for="gender">Gender</label>
         <select id="gender" name="gender" required>
            <option value="">-- Select Gender --</option>
            <option value="male">Male</option>
            <option value="female">Female</option>
            <option value="other">Other</option>
         </select>
                  <br> <br>


         <!-- Department -->
         <label for="department">Department</label>
         <select id="department" name="department" required>
            <option value="">-- Select Department --</option>
            <option value="AI-DS">AI & Data Science</option>
            <option value="CSE">Computer Science Engineering</option>
            <option value="ECE">Electronics and Communication</option>
            <option value="EEE">Electrical and Electronics</option>
            <option value="IT">Information Technology</option>
            <option value="MECH">Mechanical Engineering</option>
         </select>
                  <br> <br>


         <!-- Address -->
         <label for="address">Address</label>
         <textarea id="address" name="address"
                  rows="4"
                  placeholder="Enter your address"
                  required></textarea>
                   <br> <br>

                   <!--Submit and reset-->
```

```
            <input type="submit" value="Register">
            <input type="reset" value="Clear">
        <br> <br>

        </form>

    </div>

</body>
</html>
```

OUTPUT:

Student Registration

Fill in your details to register

Full Name

Enter your full name

Email Address

Enter your email

Password

Minimum 8 characters

Confirm Password

Re-enter your password

Mobile Number

Enter 10-digit mobile number

Date of Birth

dd-mm-yyyy

Gender

-- Select Gender --

Department

-- Select Department --

Address

Enter your address

Register Clear

CSS FILE CODE:

```
body {
    margin: 0;
    padding: 30px 15px;
    font-family: Arial, sans-serif;
    background: linear-gradient(135deg, #74ebd5, #9face6);
    min-height: 100vh;
    box-sizing: border-box;
}

/* Registration Form Container */
.form-container {
    max-width: 500px;
    margin: 20px auto;
    padding: 30px;
    background-color: white;
    border-radius: 15px;
    box-shadow: 0 8px 25px rgba(0, 0, 0, 0.2);
    animation: slideIn 1s ease-out;
}

/* Main Heading */
h1 {
    text-align: center;
    color: #34495e;
    font-size: 28px;
    margin-bottom: 10px;
}

/* Description */
.form-container p {
    text-align: center;
    color: #777;
    margin-bottom: 25px;
}

/* Labels */
label {
    display: block;
    margin-top: 12px;
    margin-bottom: 6px;
    font-weight: bold;
    color: #333;
}
```

```
/* Input Fields */
input[type="text"],
input[type="email"],
input[type="password"],
input[type="tel"],
input[type="date"],
select,
textarea {
    width: 100%;
    padding: 12px;
    margin-bottom: 5px;
    border: 1px solid #ccc;
    border-radius: 6px;
    font-size: 15px;
    font-family: Arial, sans-serif;
    box-sizing: border-box;
    transition: border-color 0.3s, box-shadow 0.3s;
}

/* Input Focus Effect */
input:focus,
select:focus,
textarea:focus {
    outline: none;
    border-color: #5967d8;
    box-shadow: 0 0 7px rgba(89, 103, 216, 0.4);
}

/* Text Area */
textarea {
    resize: vertical;
}

/* Buttons */
input[type="submit"],
input[type="reset"] {
    width: 48%;
    padding: 12px;
    margin-top: 15px;
    border: none;
    border-radius: 6px;
    background-color: #5967d8;
    color: white;
    font-size: 16px;
    cursor: pointer;
    transition: background-color 0.3s, transform 0.3s;
}
```

```
/* Button Hover Effect */
input[type="submit"]:hover {
    background-color: #3846b8;
    transform: translateY(-3px);
}

input[type="reset"] {
    background-color: #777;
}

input[type="reset"]:hover {
    background-color: #555;
    transform: translateY(-3px);
}

/* Form Opening Animation */
@keyframes slideIn {
    from {
        opacity: 0;
        transform: translateY(40px);
    }

    to {
        opacity: 1;
        transform: translateY(0);
    }
}

/* Responsive Design */
@media (max-width: 550px) {
    .form-container {
        padding: 20px;
        margin: 5px auto;
    }

    h1 {
        font-size: 24px;
    }

    input[type="submit"],
    input[type="reset"] {
        width: 100%;
    }
}
```

RESULT:

The student registration form is designed and tested using HTML5 and CSS3, incorporating browser based form validation and CSS animations.